FORM 4 CHEMISTRY

PAPER 1 (233/1)

TIME: 2 HOURS

INSTRUCTIONS TO CANDIDATES

- *Write your **Name**, **Index Number** and **School** in the spaces provided above.*
- *Answer **all** the questions in the spaces provided after each question.*
- ***KNEC** Mathematical tables and **silent non-programmable** electronic calculators may be used.*
- ***ALL** working **must** be clearly shown where necessary.*
- *Candidate should **check** the question paper to ascertain that **all** the pages are printed and that no questions are missing.*
- *Candidates* should answer the questions in **English**.

FOR EXAMINER'S USE ONLY

QUESTIONS	MAX SCORE	CANDIDATE'S SCORE
1 – 29	80	

This paper consists of 15 printed pages. Candidates should check to ascertain that all pages are printed as indicated and that no questions are missing.

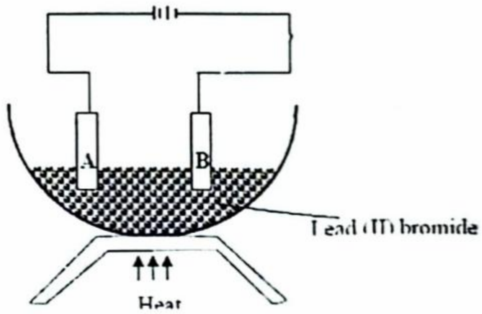
1. (a) Give the name of the first member of the alkyne homologous series (1 mark)

Ethyne.

(b) Describe a chemical test that can be used to distinguish ethanol from ethanoic acid. (2 marks)

Add sodium carbonate/sodium hydrogen carbonate solution to both ethanol and ethanoic acid, the test tube containing ethanoic acid will produce effervescence of CO_2. 01 01

2. Study the diagram below and use it to answer the questions that follow:-



(a) Identify electrodes **A** and **B** **(2mks)**

A – Anode
B – Cathode

(b) Name the product formed at the anode (1mk)

Bromine gas

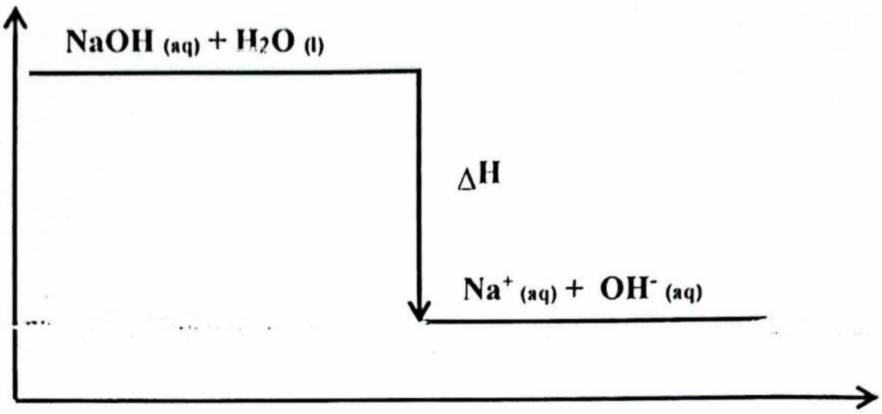
(c) Write the electrode half equation of reaction at electrode **A** **(1mk)**

$2Br^-_{(aq)} \longrightarrow Br_{2(g)} + 2e^-$

3. (a) What is meant by lattice energy? (1 mark)

Is the energy change that occurs when one mole of ionic compound is formed from its constituent ions in gaseous state.

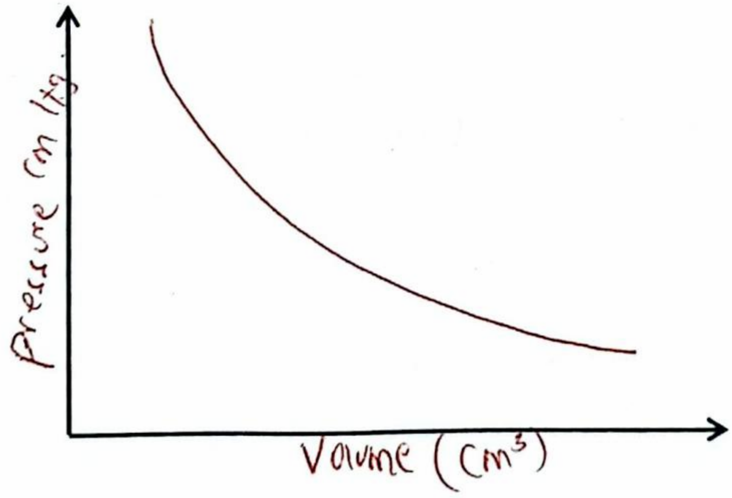
(b)Study the energy level diagram below and answer the question that follows:



What type of reaction is represented by the diagram? (1 mark)

Exothermic reaction

4. (a) Sketch a graphical representation of Boyles law on the axes below. (1 mark)



(c) A gas occupies 400 cm^3 at 25°C and 100,000 Pa. What will be its volume at 27°C and 101325 Pa? (2 marks)

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\left(\frac{100{,}000 \times 400}{298}\right) = \left(\frac{101325 \times x}{300}\right)$$

$$\frac{337.75x}{337.75} = \frac{134{,}228.1879}{337.74}$$

$$= 397.42\,cm^3$$

5. (a) What is half- life? (1 mark)

is the time taken for a given mass or number of nuclides to decay to half its original mass or number.

(d) The half-life of protactinium - 234 is 1.17 minutes. Determine the mass that decays in 5.85 minutes starting with 100 g of the sample. (2 marks)

$n = \frac{\text{total time}}{\text{half-life}}$ | Remaining amount $= \left(\frac{1}{2}\right)^n \times$ original amount

$= \frac{5.85}{1.17} = 5.24$ | $x = \left(\frac{1}{2}\right)^{5.24} \times 100$ | $x = 2.64\,g$

6. State **two** disadvantages of hard water. (2 marks)

(i) Stains cloths ✓ 1

(ii) Waste soap ✓ 01

(iii) Deposition of fur (calcium carbonate) in kettles, pipes & boilers.

7. Hydrogen chloride gas can be prepared by reacting sodium chloride with an acid.

(a) Name the acid. (1 mark)

Conc. Sulphuric (VI) acid.

(a) Write an equation for the reaction between sodium chloride and the acid. (1 mark)

$NaCl_{(aq)} + H_2CO_4 \longrightarrow Na_2SO_4 + HCl_{(g)} +$

(c) State **two** uses of hydrogen chloride. (1 mark)

(i) used in manufacture of hydrochloric acid.

(ii) used in manufacture of various chemicals and products.

8. When solid **B** was heated strongly, it gave off water and a solid residue. When water was added to the solid residue, the original solid **B**, was formed.

(a) What name is given to the process described? (1 mark)

Reversible reaction

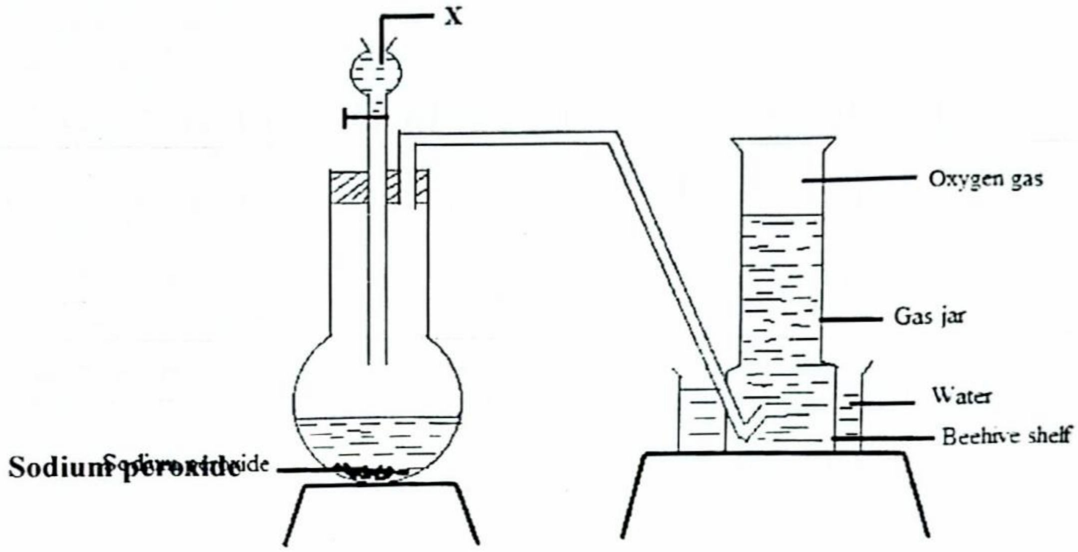
....

(b) Give **one** example of solid **A**. (1 mark)

....

Hydrated Copper (II) Sulphate.

9. The setup below can be used to prepare oxygen gas. Study it and answer the questions that follow.



(a) Identify **X**. (1 mark)

Water.

(c) Write the equation for the reaction which occurs in the flask. (1 mark)

$Na_2O_2{}_{(s)} + H_2O_{(l)} \longrightarrow 2NaOH_{(aq)} + O_2{}_{(g)}$

(d) State **one** use of oxygen other than in welding (1 mark)

i) used in hospital ~~with~~ by people with breathing difficulties.
ii) used by Mountain climbers and deep sea divers.

10. The atomic number of an element, **M** is 13.

(a) Write the electronic configuration of the ion M^{3+}. (1 mark)

2.8.8

(b) Write the formula of the chloride of **M**. (1 mark)

MCl_3

MCl_3

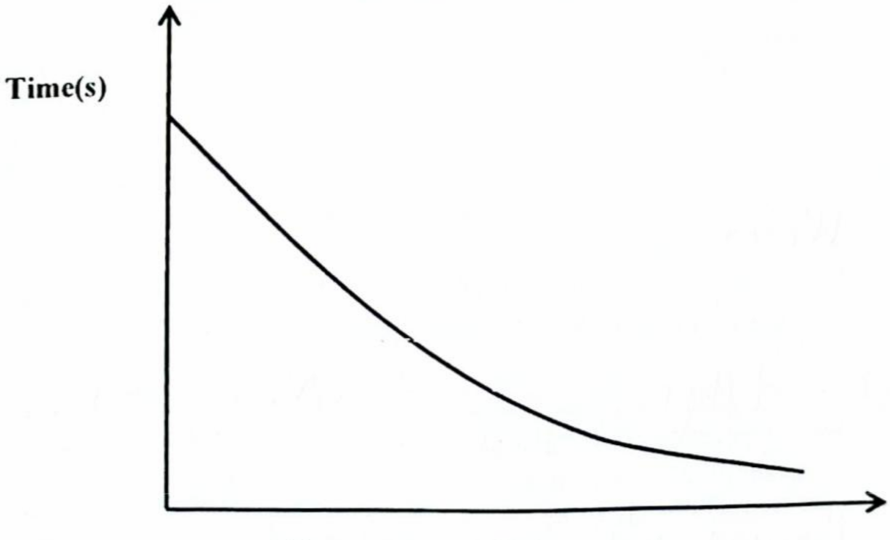
(c) State the structure of the compound formed in (b) above (1 mark)

ionic structure.

11. Concentrated sodium chloride was electrolysed using graphite electrodes. Name the product formed at the anode and give a reason for your answer. (2 marks)

Chlorine gas -> This is because chloride ion is discharged because of its high concentration, it looses electrons to form chlorine gas.

12. The curve shown below shows the variation of time against temperature for the reaction between sodium thiosulphate and hydrochloric acid.



(a) Explain the shape of the curve. (2 marks)

The rate of reaction is inversely proportional to time taken and directly proportional to temperature.

(b) Other than temperature name **one** factor that affects the rate of reaction. (1 mark)

(i.) Pressure · in

(ii) Concentration ·

13. (a) Dry ammonia was passed over heated copper (II) oxide in a combustion tube.

(i) State the observations made in the tube (1 mark)

Black copper (II) Oxide turns to brown copper metal.

(ii) Write an equation for the reaction that occurs. (1 mark)

$3CuO_{(s)} + 2NH_{3(g)} \longrightarrow 3Cu_{(s)} + N_{2(g)} + 3H_2O_{(g)}$

(e) What products would be formed if red hot platinum is introduced into a mixture of ammonia and oxygen? (1 mark)

Nitrogen (II) oxide and Steam

14. The table below shows behaviour of metals P, Q, R and S. Study it and answer the questions that follow:

Metal	Appearance on exposure to air	Reaction with water	Reaction with dilute sulphuric (VI) acid
P	Remains the same	Doesn't react	Reacts moderately
Q	Remains the same	No reaction	Doesn't react
R	Slowly tarnishes	Slow	Vigorous
S	Slowly turns white	Vigorous	Violent

(a) Arrange the metals in the order of reactivity starting with the most reactive. (2 marks)

S R P Q

$\longrightarrow$

Decreasing reactivity.

(f) Name a metal which is likely to be **R** (1 mark)

....

.........Magnesium...

15. Given the following substances: sodium carbonate, orange juice and sodium bromide.

(a) Name **one** commercial indicator that can be used to show whether sodium carbonate, orange juice and sodium bromide are acidic, basic or neutral. (1 mark)

.........Litmus paper...

.........Phenolphthalein indicator...

(b) Classify the substances in 15 (a) above as acids, bases or neutral. (2 marks)

Acid	orange juice
Base	
Neutral	sodium bromide, sodium carbonate

16. (a) One of the allotropes of sulphur is monoclinic sulphur, name the other allotrope (1 mark)

.........Rhombic sulphur...

(g) Concentrated sulphuric (VI) acid reacts with copper and prapanol. State the property of the acid shown in each case. (2 marks)

Copper ....Oxidizing agent...

Propanol ....Dehydrating agent...

17. Study the standard electrode potentials in the table below and answer the questions that follow.

Half -reaction	E^0(V)
$Ag^+(aq) + e \longrightarrow Ag(s)$	+ 0.80
$Cu^{2+}(aq) + 2e \longrightarrow Cu(s)$	+ 0.34
$Mg^{2+}(aq) + 2e \longrightarrow Mg(s)$	- 2.38
$Ca^{2+}(aq) + 2e \longrightarrow Ca(s)$	- 2.87

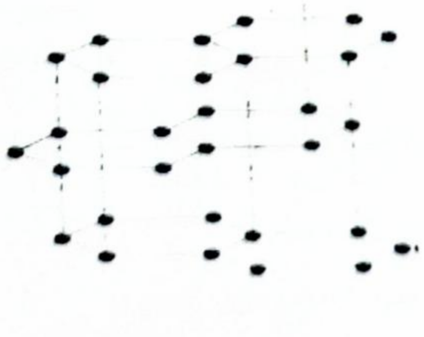
(a) Which of the metals is the strongest oxidising agent? (1 mark)

Ag.

(b) What observations will be made if a copper coin was dropped into an aqueous solution of calcium nitrate? Explain. (2 marks)

No change because copper is less reactive than calcium and hence cannot displace calcium from its ion.

18. The diagram below represents an allotrope of carbon.



a) name the allotrope. (1mk)

Graphite

b)Explain why:- (2mks)

(i) its slipperly

Its atoms are held by weak van der waals forces which can slide over each Other.

(ii) Conducts an electric current

uses three out of four electrons for bonding. one electron is left delocalised hence conduct electricity.

19. (a) A crystal of iodine, heated gently in a test tube gave off a purple vapour.

(i) Write the equation for the formation of purple vapour. (1 mark)

$I_{2(s)} \longrightarrow I_{2(g)}$

(b) What type of bond is broken when the iodine crystal is heated gently? (1 mark)

Van der waals force.

(h) State **one** use of chlorine. (1 mark)

(i) Chlorine is used to make bleache.
(ii) used to kill Micro-organisms in water treatment works.

20. Describe how samples of barium (II) sulphate, ammonium chloride and common salt can be obtained from a mixture of the three. (3 marks)

-> Heat the mixture and ammonium chloride sublimes leaving barium sulphate and Common Salt
-> Add water,

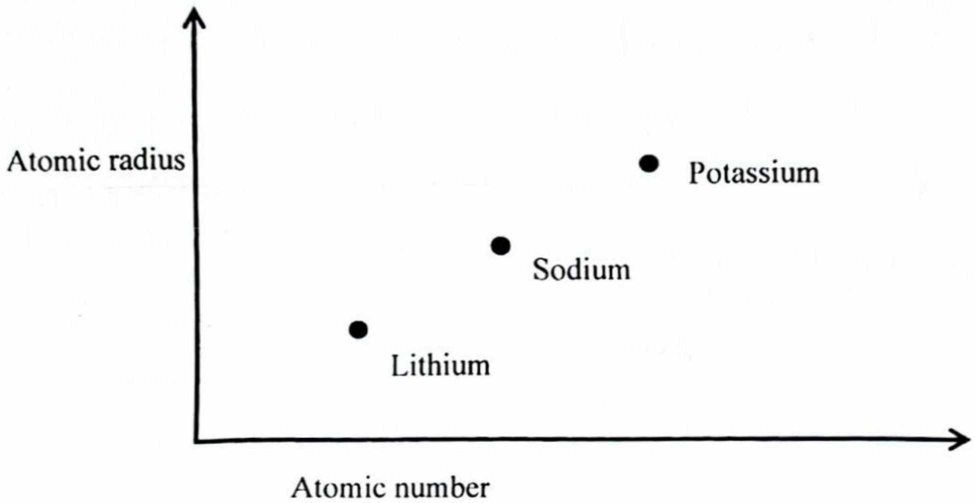
21. (a) Give the name of the process which takes places place when maize flour is converted to ethanol (1 mark)

Fermentation

(i) Write the formula of the compound formed when ethanol reacts with sodium metal. (1 mark)

$CH_3CH_2OH_{(aq)} + Na_{(s)} \longrightarrow CH_3CH_2ONa + H_{2(g)}$

22. (a) Study the graph below which shows variation of atomic radius with atomic number



State and explain the trend shown in the graph above. (2 marks)

The number of energy levels increases by one from Lithium to Potassium down the group. This is due to increase in the number of electrons. Atomic number is proportional to the number of electrons.

(j) State **one** use of sodium. (1 mark)

✓ Sodium is used in the Manufacture of Sodium Cyanide for use in extraction of gold.
✓ Sodium Chloride is used as a food additive.

23. A farmer intended to plant blueberries in her farm. She first tested the pH of the soil and found it to be 10.0. In order to obtain high yield, what advice would be given to the farmer if blueberries do well in acidic solution? (2 marks)

Adding acidifying fertilizers e.g. Ammonium sulphate, urea, ammonium nitrate. Nitrogen sources create acidity as they break down

24. Starting with calcium nitrate solution, describe how a pure dry sample of calcium carbonate can be prepared in the laboratory. (3 marks)

Put the specific amount of Calcium nitrate solution in the beaker. Add excess amount of Sodium carbonate solution and mix. The cations will exchange anions. Calcium carbonate solid and Sodium nitrate solution are formed. Decant to obtain Calcium carbonate residue and wash using distilled water. then dry between the filter papers.

25. A hydrocarbon contains ~~81.82%~~ 85.71 of carbon. If the molar mass of the hydrocarbon is ~~44~~ 42, determine the molecular formula of the hydrocarbon. (C = 12.0; H = 1.0) (3 marks)

$C = \frac{85.71}{12} = \frac{7.1425}{7.1425} = 1$, $H = \frac{14.29}{1} = \frac{14.29}{7.1425} = 2$

$n(CH_2) = 42$

$\frac{n(14)}{14} = 3$

$n(CH_2)$

$3(CH_2) = C_3H_6$

26. (a) Describe how Carbon (II) Oxide can be distinguished from Carbon (IV) Oxide using calcium hydroxide solution. (2 marks)

Bubble the two gases through Calcium hydroxide Solution. The Carbon (IV) oxide forms white precipitate with lime water while Carbon (II) oxide does not have effect on Calcium hydroxide

(k) What is the role of carbon (IV) oxide in fire extinguishing? (1 mark)

It forms a blanket cutting of the supply of oxygen since it is denser than air.

27. (a) Name **one** source of alkanes. (1 mark)

Crude oil, biogas and natural gas.

(I) Methane gas was reacted with one mole of chlorine gas. State the condition necessary for this reaction. (1 mark)

Presence of sunlight / UV-light.

28. (a) What is meant by heating value of a fuel? (1 mark)

The amount of heat energy given out when a unit mass or unit volume of a fuel is completely burned in oxygen.

(b) Other than heating value, name **one** factor to be considered when choosing a fuel. (1 mark)

- Ease and rate of combustion
- Cost
- Ease of storage
- Availability
- Ease of transportation
- Environmental effects

29. The following diagram shows a paper chromatogram of substances A, B, C, and D which are coloured